01.06.2021 - Εργαστήριο

```
%% ASKHSH 3
%% POSOTHTES POY PERIGRAFONTAI ME EKQETIKES SYNARTHSEIS, DHLADH SYNARTHSEIS
%% THS MORFHS y = a * exp(b*x);
%% YPOLOGISTE THN EKQETIKH SYNARTHSH POY PERIGRAFEI TA SHMEIA x KAI y,
%% KANTE TH GRAFIKH PARASTASH STO IDIO GRAFIKO PARAQYRO.
%% EPISHS YPOLOGISTE TO SXETIKO SFALMA THS PROSEGGISHS.y

pause; figure(1); clf; clear all; disp('ASKHSH 3');
x = [1 2 3 4 5 6 7 8];
y = [0.74 0.20 0.065 0.019 0.0053 0.0018 0.0006 0.0002];
plot(x, y, 'k*'); hold on;

logy = log(y);
synt = polyfit(x, logy, 1);

a = exp(synt(2))
b = synt(1)
xplot = [1:0.1:8];
yplot = a*exp(b*xplot);
plot(xplot, yplot)

sxetikosfalma = norm(y - a*exp(b*x)) / norm(y)
```

$y = a e^{\beta x} \Leftrightarrow \ln(y) = \ln(a) + \beta x$

$\boxed{Y = A + \beta x}$

πολυώνυμο 1ου βαθμού (polyfit, 1ου βαθ.)

δεδομένα

Y

$synt = \begin{bmatrix} \beta \\ A \end{bmatrix}$

$A = \ln(a) \Leftrightarrow \boxed{a = e^{A}}$

$$\rightarrow = \frac{\| \underline{y} - f(\underline{x}) \|_2}{\| \underline{y} \|_2}$$

$$\int_0^{10} h = ?$$

```
clf; clc; clear all, disp('ASKHSH 2')
h = inline('1./((x-0.3).^2+0.01)  + 1./((x-0.9).^2+0.4)');
ezplot(h,[0,10]); title('1/((x-0.3)^2+0.01)  + 1/((x-0.9)^2+0.4)'); axis tight;
x = [0:0.5:10];
y = h(x);
olokt = trapz(x,y);
fprintf('TRAPZ: To oloklhrwma ths h sto [0, 10] einai %12.8f \n',olokt);
hold on;
plot(x,y, '*-r')
```

Βάζω τις h

Προσαρμόζω το γραφικό παράθυρο ακριβώς στις τιμές της γραφικής παράστασης

$$h(x) = \frac{1}{(x-0{,}3)^2+0.01} + \frac{1}{(x-0.9)^2+0.4}$$

Κάλεσα τις trapz (σύνθετος τύπος).

```
%% ASKHSH 1
%% Symbolikoi ypologismoi gia ypologismo oloklhrwmatwn se synexh dedomena.
%% Oloklhrwste th synarthsh 1 metablhths f(x) = 1./sqrt(1+x.^4)

clc; clear all, disp('ASKHSH 1')
ekf1 = '1./sqrt(1+x.^4)';
f1 = inline(ekf1);
ezplot(f1, [-2, 2]);
olok1 = quad(f1, -2, 2);
fprintf('QUAD: To oloklhrwma ths synarthshs %s einai %12.8f \n',ekf1,olok1);
```

$$f_1(x) = \frac{1}{\sqrt{1+x^4}}$$

υπολογίζω το ολοκλήρωμα της f1.